Community Robotics Workshops

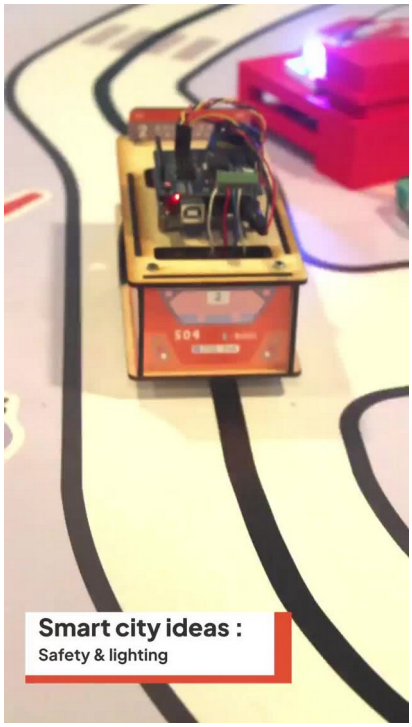
Aurinkolab | Project concept for community partners | October 2026

We help communities run robotics workshops and line-following competitions with reusable kits, learning materials and remote facilitator training. A pilot provides 10 robots for 10–30 participants: one kit per individual or team of up to three. Participants develop coding skills, teamwork and a connection to their city.

Built on experience in Riihimäki

In our Riihimäki project, students explored their city through robotics, designed circuits and 3D models, and used digital fabrication to develop a city-map track and an R-bussi robot. Partners included International House Riihimäki, Robotics Campus and Hyria, supporting the project, school cooperation and production respectively. See the project video.





Riihimäki in practice: participants explore their city through making, and the R-bussi follows the track. Stills from Dmitrii Spivak's project video.

What communities receive

Ten Arduino-compatible robot kits, each with two motors, up to four line sensors, wheels, a bus body, a disposable 9 V PP3 battery, holder, cables and fasteners; one shared printed track; spare parts; existing open-source assembly instructions, code examples and competition rules, provided at no additional preparation charge. Participants use mBlock for block-based programming or C++ with Arduino Wiring. Tasks progress from simple turning rules to comparing sensor layouts and P/PD/PID control using analogue readings.

Local delivery and lasting value

The 1–2 day event covers assembly, programming and timed runs. We provide eight hours of remote facilitator training and consultation. Local facilitators teach participants; the host supplies a venue, laptops and tools. Kits stay with the community. Reporting covers attendance, completed robots and feedback.



The Riihimäki city-map track. The optional €1,000 adaptation can feature local landmarks and robot artwork for another city or region.

Indicative pilot budget

10 complete kits with disposable 9 V batteries — 10 × €124.44	€1,244.40
Spare parts/batteries €130; track €100; packing/delivery €120	€350.00
Kit preparation 4 h × €50; facilitator training/support 8 h × €87	€896.00
Contingency — 15% of the base project	€373.56
Base project total	**€2,863.96**
Optional city/region customisation of the map and robot designs	+ €1,000.00

Planning estimate, not a supplier quotation: purchases include retail VAT without recovery; labour uses €50/h for kit preparation and €87/h for training/support; no margin is included. Disposable 9 V battery operation and runtime require a prototype test. Local staffing, venue and laptops are excluded. Customisation is optional.

Timing and expansion

Delivery is three months from agreement on scope, kit specification and artwork. Optional city/region customisation brings the total to €3,863.96; no extra contingency is added to the option. Oulu is a proposed pilot location; Dubai requires local partners and a separate delivery budget.

Contact: Dmitrii Spivak • LinkedIn